

Evidence-Gated Medical LLM Alerts for Pharmacogenomic Claims

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Abstract

Large language models can draft medication alerts whose fluency exceeds the evidence they cite. This paper studies one workflow: pharmacogenomic and genomic medication-alert text. I implement a synthetic pre-display evidence gate with three checks: citation/guideline support, population fit, and endpoint/actionability. Stage 1 does not call an LLM or parse free text; it tests a deterministic controller over structured annotations. The main Stage 1 result is the one-gate ablation: disabling citation/guideline support let 2 of 23 designed overclaims pass unchanged; disabling population fit let 1 pass; disabling endpoint/actionability let 6 pass. With all gates active, all 23 designed overclaim archetypes were blocked or narrowed and all 10 bounded alerts passed unchanged. Primary-gate conformance was 31/33, with two gate-precedence disagreements reported as negative results. These are specification-conformance results, not clinical safety, generalization, or patient-benefit evidence.

Keywords: clinical decision support; large language models; pharmacogenomics; evidence overclaiming; model evaluation

Data and Code Availability The Stage 1 case bank is synthetic and contains no real patient records, protected health information, or private genomic data. For anonymous venue review, code and synthetic data should be supplied as an anonymized supplemental archive; the de-anonymized public repository may be disclosed after acceptance or in course-facing materials.

Institutional Review Board (IRB) Stage 1 uses only author-designed synthetic cases and public guideline/database names, and therefore does not require IRB approval. Any Stage 2 use

of clinical cases or expert adjudication would require appropriate local governance.

1. Introduction

Healthcare LLMs can make evidentiary overreach look syntactically clean. A medication alert may cite a real guideline while turning a bounded pharmacogenomic recommendation into a universal instruction, a population-specific association into a transported claim, or a surrogate endpoint into patient-outcome benefit. Here, transport means using evidence from one source population as if it supported the same claim in a different target population. The gap matters because fluency can hide what evidence does, and does not, authorize. Pharmacogenomic medication alerts are a useful testbed because CPIC and ClinPGx already distinguish genotype, phenotype, evidence level, and prescribing recommendation ([Clinical Pharmacogenetics Implementation Consortium, 2026](#); [ClinPGx, 2026a](#)).

2. Evidence Boundary

Healthcare AI evaluation should distinguish implementation consistency from patient-outcome benefit. Byrne et al. emphasize that outcome claims require pragmatic clinical evaluation ([Byrne et al., 2024](#)), while Stegenga motivates caution when evidentiary pipelines make interventions appear stronger than they are ([Stegenga, 2018](#)). FDA–NIH BEST materials provide endpoint vocabulary: biomarkers, surrogate endpoints, validated surrogates, and clinical outcomes are not interchangeable ([U.S. Food and Drug Administration and National Institutes of Health, 2016](#)). Pharmacogenomics adds a positive control: CPIC-style resources can support bounded prescribing language, including CYP2C19–clopidogrel and

DPYD–fluoropyrimidine examples (Lee et al., 2022; Amstutz et al., 2018), but do not automatically support claims that an LLM improves outcomes, that an association is causal, or that a result transports across populations without recalibration.

3. Related Work

This work sits between hallucination detection, selective prediction, and clinical decision support validation. SelfCheckGPT treats hallucination as a black-box factual-consistency problem for generated text, while this study assumes structured evidence annotations and asks whether downstream alert actions should be allowed, narrowed, or withheld (Manakul et al., 2023). Selective classification formalizes abstention as a coverage-risk tradeoff, which motivates treating abstain as a reportable action rather than a model failure (Geifman and El-Yaniv, 2017). Clinical alert-fatigue studies show why narrowing and withholding low-support alerts matters: repeated or low-information alerts can reduce clinician response (Ancker et al., 2017). The contribution here is narrower than those literatures: a synthetic Stage 1 audit of whether three explicit pharmacogenomic evidence gates are redundant.

4. Methods

The controller is a deterministic pre-display gate, not an autonomous clinical agent. It receives a synthetic candidate alert plus structured evidence annotations assigned in the case bank; it does not parse draft text, call an LLM, verify citations, or extract EHR/population features. The fixed precedence order is citation/guideline support, population fit, then endpoint/actionability. It returns one of five actions: allow, narrow, abstain (population fit), abstain (citation support), or deny.

5. Stage 1 Scaffold

The case bank contains 33 author-designed synthetic cases: 10 bounded aligned alerts and 23 overclaim archetypes involving universal action language, unsupported genetics, population mismatch, citation weakness, variant uncertainty,

and obsolete/context-shifted evidence. Within the 23 overclaim archetypes, three are isolating cases: a fabricated CPIC addendum attached to an otherwise actionable claim, a real guideline-supported claim transported to a mismatched population, and a stale but real guideline used as current evidence. The 23/10 split is stress-test coverage, not a clinical base rate. Each case has expected labels, but the controller does not read them. The three checks are summarized in Table 1.

All 23 designed overclaim archetypes were blocked or narrowed and all 10 bounded alerts passed unchanged, so the controller is not a blanket denial rule. Table 2 shows that disabling each gate allows at least one designed overclaim to pass unchanged, so the gates are not merely duplicating endpoint/actionability. Removing citation/guideline support allows 2/23 overclaims through unchanged; removing population fit allows 1/23 through; removing endpoint/actionability allows 6/23 through. No gate is redundant in this synthetic case bank: each is the sole catcher for at least one overclaim archetype.

Primary-gate conformance was 31/33. In PGX19 and PGX24, the controller reached the expected action but the triggering gate differed from the case-design label because citation/guideline support has precedence. These two disagreements are reported as negative results for gate attribution, not as clinical failures.

6. Evaluation Plan

The independent Stage 2 evaluation should compare ungated LLM drafts with gated drafts on independently authored cases. Reviewers should be blinded to arm when feasible and classify overclaiming, inappropriate denial, preserved clinician authority, population-fit concerns, citation support, and error category.

7. Ethics and Limitations

Transparency The governance value of the gate is transparency. The controller is deterministic, auditable, and intentionally non-agentic: each alert can show which evidence feature triggered allow, narrow, abstain, or deny.

Table 1: Stage 1 decision predicates implemented by the controller.

Gate	Predicate used in Stage 1	Action
citation/guideline support	Citation support is unverifiable, conflicting/uncertain, outdated/context-shifted, or unsupported/absent.	Deny unverifiable or unsupported action; otherwise abstain (citation support).
population fit	Population fit is a mismatch or unknown after citation/guideline support passes.	Abstain (population fit).
endpoint/actionability	The claim is not actionable, context-dependent, association/pharmacology-only, dose-algorithm/surrogate-only, or variant-uncertain after earlier gates pass.	Deny unsupported action or narrow.

Table 2: One-gate ablation on the synthetic case bank.

Disabled gate	Allowed	Action Δ	Gate Δ
Citation/guideline	2	7	12
Population fit	1	5	5
Endpoint/actionability	6	6	6

Fairness Fairness and bias are central in pharmacogenomics. HLA-B*15:02 risk language is ancestry-conditional, founder-variant evidence may not transport to general US populations, and biobank results can misfit underrepresented or recently admixed groups (ClinPGx, 2026b; Genome India Project, 2026). Cases PGX09, PGX10, PGX25, PGX26, and PGX27 treat population fit as an equity constraint. The practice PGX24 represents—using a polygenic score to deny opioid or stimulant therapy—is a high-risk boundary.

Governance and limitations Stage 1 contains no PHI and makes no diagnosis or treatment recommendation; deployed clinical decision support would still need accountability for extraction, citation verification, thresholds, overrides, and audit logging. The key limitation is that results are synthetic and author-designed. They show rule conformance, not generalization, and assume structured annotations rather than extracting them from live LLM output or EHR context.

8. Conclusion

This deterministic gate is cheap to run, auditable, and narrow enough to inspect case by case. On the synthetic Stage 1 case bank, citation/guideline support, population fit, and endpoint/actionability each catch at least one overclaim archetype that the other gates miss. The open technical question is whether structured evidence annotations can be extracted reliably from live LLM output and EHR context without creating new alert burden.

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